

KOLEJ UNIVERSITI TUNKU ABDUL RAHMAN
FACULTY OF COMPUTING AND INFORMATION TECHNOLOGY
ACADEMIC YEAR 2021/2022
OCTOBER EXAMINATION
COMPUTER SCIENCE BACS2163
SOFTWARE ENGINEERING

MONDAY, 4 OCTOBER 2021

TIME: 9.00 AM – 12.00 NOON (3 HOURS)

BACHELOR OF COMPUTER SCIENCE (HONOURS) IN DATA SCIENCE
BACHELOR OF COMPUTER SCIENCE (HONOURS) IN SOFTWARE ENGINEERING

BACHELOR OF INFORMATION TECHNOLOGY (HONOURS) IN SOFTWARE SYSTEMS
DEVELOPMENT

Instructions to Candidates:

Answer **ALL** questions in the requested format or template provided.

- This is an open book final online assessment. You **MUST** answer the assessment questions on your own without any assistance from other persons.
- You must submit your answers within the following time frame allowed for this online assessment:
 - The deadline for the submission of your answers is **half an hour** from the end time of this online assessment.
- Penalty as below **WILL BE IMPOSED** on students who submit their answers late as follows:
 - The final marks of this online assessment will be reduced by 10 marks for answer scripts that are submitted within 30 minutes after the deadline for the submission of answers for this online assessment.
 - The final marks of this online assessment will be downgraded to zero (0) mark for any answer scripts that are submitted after one hour from the end time of this online assessment.
- Extenuation Mitigating Circumstance (EMC) encountered, if any, must be submitted to the Faculty/Branch/Centre within 48 hours after the date of this online assessment. All EMC applications must be supported with valid reasons and evidence. The UC EMC Guidelines apply.

FOCS Additional Instructions to Candidates:

- Include your **FULL NAME**, **STUDENT ID** and **PROGRAMME OF STUDY** in your submission of answer.
- Read all the questions carefully and understand what you are being asked to answer.
- Marks are awarded for your own (original) analysis. Therefore, use the time and information to build well-constructed answers.

This question paper consists of 4 questions on 4 printed pages.

BACS2163 SOFTWARE ENGINEERING**Question 1**

Wow Fashion Academy is a newly established institution in Kuala Lumpur which offers make-up, hairstyling, and art courses. The courses are conducted in both full time and part time modes. Recently, the academy plans to switch to online platforms due to COVID-19 pandemic and has contacted your company as a suitable software house to handle this project.

You, as the project manager of the project is responsible to build an online learning system with the core functions for the academy to allow customers to register as member, pay the fees, check classes' schedule, attend classes online and download materials in 3 months. Furthermore, the academy also wishes to extend their online system to share success story, subscribe to weekly newsletter, chat with company personnel, and connect to other social networks such as Facebook Page, Twitter, and Instagram in the next version.

The top management, the academy staff, the administration staff and the students are very supportive of this upcoming project. However, there are only two software engineers available to assist you in this project.

- a) Identify if the online learning system as mentioned above is a generic software product or bespoke software product? Justify your answer with **TWO (2)** reasons. (1 + 4 marks)
- b) Recommend and explain **ONE (1)** software process model to develop the online learning system as mentioned above. Justify your recommendation with **FOUR (4)** reasons. (2 + 8 marks)
- c) Suggest and explain **THREE (3)** software quality attributes for this online learning system as mentioned above. Justify your suggestions. (6 marks)
- d) As a project manager, you would need to consider the factors that could motivate employees throughout the project. Identify and explain **TWO (2)** motivating factors for this case based on the Maslow Motivation Model. (4 marks)

[Total: 25 marks]

Question 2

Rapid Grocery is a mid-sized family business that operates since 30 years ago. The main operations in the grocery store such as stock in and stock out, are still performed manually by the staff. To ease the daily operations, the owner of Rapid Grocery wants to computerize the stock in and stock out operations in the store. In addition, he also wants to enable online ordering services to the customers. Nevertheless, the owner does not have any background or knowledge in computerized and online systems. He has contacted your company, a software house, to handle this project.

- a) Analyse and briefly explain **TWO (2)** techniques for discovering requirements for the above project. (4 marks)
- b) Analyse **THREE (3)** functional requirements from the above online ordering system and **TWO (2)** non-functional requirements. (5 marks)

BACS2163 SOFTWARE ENGINEERING**Question 2 (continued)**

- c) Prepare a Gantt Chart to develop the online ordering system based on the following table. Assume that the project starts in July 2021. (9 marks)

Task ID	Task Description	Predecessor	Duration (month)	Overlap (month)
A	Requirement Engineering	None	$\frac{1}{2}$	None
B	System Analysis & Design	A	$\frac{1}{2}$	$\frac{1}{4}$
C	Customer Feedback	A	$\frac{1}{2}$	$\frac{1}{4}$
D	Coding	B,C	1	$\frac{1}{4}$
E	Customer Feedback	B,C	1	$\frac{1}{4}$
F	Testing	E	$\frac{1}{2}$	None
G	Deployment	F	$\frac{1}{4}$	None

- d) Assume that you are planning to adopt a viewpoint-oriented analysis method for the requirements analysis process. Identify **FIVE (5)** viewpoints for the online ordering system. Construct a viewpoint hierarchy to present your viewpoints. (7 marks)

[Total: 25 marks]

Question 3

HotNews Press is a newspaper company which started its first printed newspaper in 1970s. The company is maintaining a computerized system with text file storage server which stores large amount of the newspaper draft contents (original story from the interview, news draft, supporting images or news, and et cetera). The file storage system has no documentation nor proper module design. The draft contents might be useful in the future to trace the original story. Recently, you as the IT project manager are invited to attend a discussion meeting with HotNews Company's IT department. The main agenda of the meeting is to decide on the maintenance of the legacy systems.

- a) Based on your experience, discuss the **TWO (2)** factors that could affect the maintenance cost of HotNews's legacy system. (4 marks)
- b) Draw a diagram to explain legacy system assessment and management based on "Business Value" and "System Quality" areas. Explain and justify which strategy you should apply to maintain HotNews's legacy system. (4 + 3 marks)
- c) Assume that HotNews IT department has decided to re-engineer the legacy system, propose **ONE (1)** system organization/structuring design model. Explain and justify your answer. (6 marks)
- d) Suggest **FOUR (4)** user interface design principles that must be applied in the HotNews's proposed system. Justify your suggestions. (8 marks)

[Total: 25 marks]

BACS2163 SOFTWARE ENGINEERING**Question 4**

FirstTech is a well-known private college who provides quality education focusing in computer science and engineering. The president of FirstTech is planning to computerize a series of services such as facilities booking, library book return and borrow, photocopying machines, and et cetera. Students' identity cards will be converted into electronic identity (e-ID) to ease the students to access the series of services provided by FirstTech. He is also planning to implement automation of lighting during night and in the washroom, office air-con automated temperature detection and adjustment, video monitoring system, and et cetera in the college.

- a) Identify and explain **TWO (2)** sets of stimuli-responses for the automation of a lighting system. (6 marks)
- b) Propose and explain **TWO (2)** functional classification of Computer Assisted Software Engineering (CASE) tools with **TWO (2)** specific examples of tools for each classification that can be used by FirstTech college. (8 marks)
- c) Plan and explain **TWO (2)** testing techniques that can be used in testing the facility booking module of FirstTech computerized services system. You must include **TWO (2)** examples of test cases for each testing technique. (11 marks)

[Total: 25 marks]